Þunresdæg v1.1.2

created by k.odk

01 · THE IDEA

Made to be played

Þunresdæg makes sound from maps — the small equations that are iterated over and over, the tent map and its relatives — and runs it through a board of effects that you patch with cables.

Nothing is sampled. Every sound is a number sequence run through the board: three map oscillators, two phasers, an analogue-style delay, a row of pedal-type effects and a 4×4 matrix mixer that lets the paths cross. The eight pads on the front are not knobs. They are control voltages, and what they move is decided on the board.

It is made to be played, a little like a modular patch that lives in your hand, after the analogue modular tradition. And nothing in it is random: the maps are deterministic, and even the RANDOM buttons step a fixed sequence — the same presses from launch give the same patch.

Too much to learn? Skip it. Open the **BOARD** and press **RANDOM** a few times — each press builds a new patch from the oscillators to OUT, with some pads already wired in. Then close it and play the pads. When you find something, press **SOFT** to shake it gently, or hold a preset number to keep it.

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02 · REQUIREMENTS

What it needs

Þunresdæg iterates three maps at audio rate and runs a patchable board of effects in real time, while it can also watch the camera. It asks more of the device than a typical app.

System

IOS	iOS 18.0 or later.
DEVICE	iPhone, played in landscape. **iPad:** the iPad layout is ready and will arrive in the next update (see section 10).
RECOMMENDED	iPhone 13 or newer. Þunresdæg is developed and tested on an iPhone 13. Lower the camera FPS if the sound starts to crackle.
CAMERA	Front or back, for camera mode: the pads follow motion or light. No image is ever stored.
MICROPHONE	Optional — the MIC box on the board brings the microphone or an external interface (EX) in.

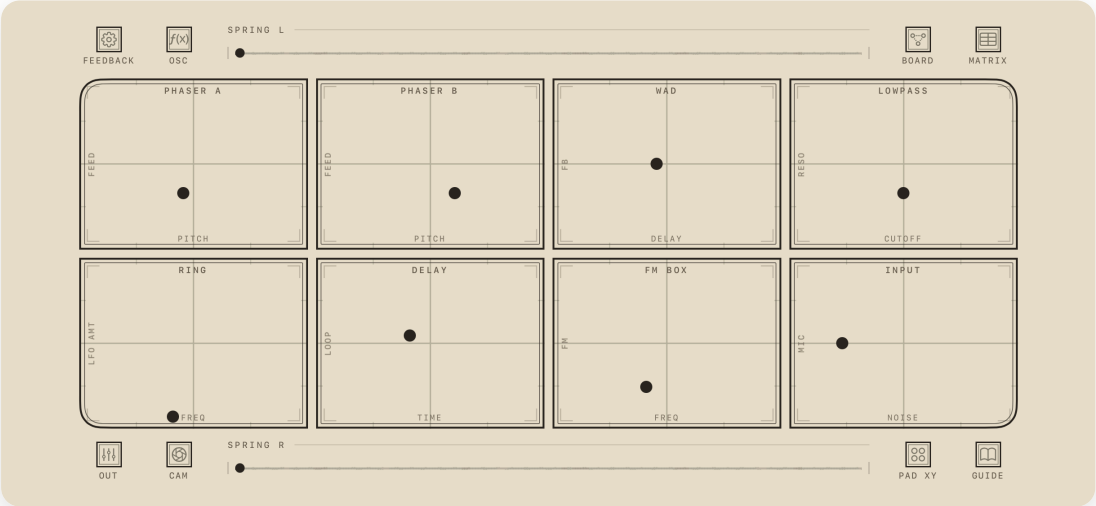
About AUv3

Þunresdæg is a **standalone app**, not an AUv3 plug-in. The camera, the microphone and the pads are part of the instrument, and a plug-in inside a host cannot use them in the same way.

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03 · THE FRONT

Eight pads, and what they move FRONT



The pads

Eight pads, four by two. They are not knobs but control voltages: X and Y of each pad go wherever they are patched on the BOARD, and one pad can move several things at once. Each pad shows what it is moving now — in the first patch **PHASER A** and **B** (PITCH, FEED), **WAD** (DELAY, FB), **LOWPASS** (CUTOFF, RESO), **RING** (FREQ, LFO AMT), **DELAY** (TIME, LOOP), **FM BOX** (FREQ, FM) and **INPUT** (NOISE, MIC). Repatch, and the names follow.

SPRING L · SPRING R

The two lines above and below the pads are the two spring reverbs. Slide them for how much of each is heard; the line trembles with the springs.

The keys

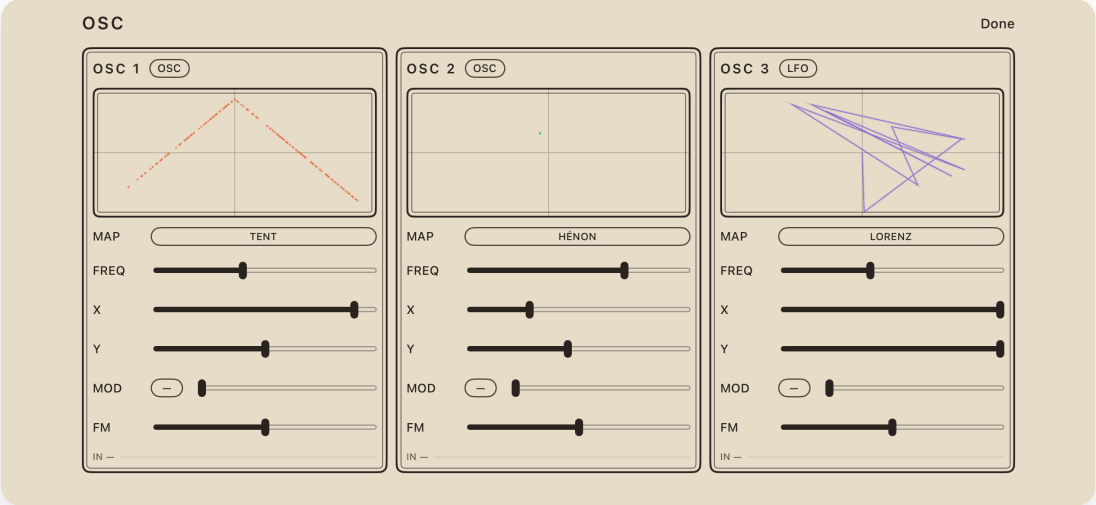
FEEDBACK	the two phasers and WAD (section 06)
OSC	the three map oscillators (04)
BOARD	cables, pads and presets (05)
MATRIX	the matrix and the springs (08)
OUT	the exit desk: mixer, EQ, compressor, limiter (09)
CAM	camera mode on and off (09)
PAD XY	the pads hold the matrix directly (08)
GUIDE	the guide inside the app

The chain

OSC 1–3 make the numbers; the **BOARD** sends them through phasers, delay and pedals; the **MATRIX** lets the paths cross; then EQ, compressor, limiter and **OUT**.

04 · OSC — THE MAP OSCILLATORS

Three equations, iterated OSC



Each oscillator steps a map at the rate **FREQ** sets and reads its state out as sound. **MAP** picks the equation — TENT, HÉNON, LORENZ and their relatives. **X** and **Y** are the map's own parameters: on the tent map X is the slope, on Hénon they are a and b, so the same slider makes a very different kind of change from map to map.

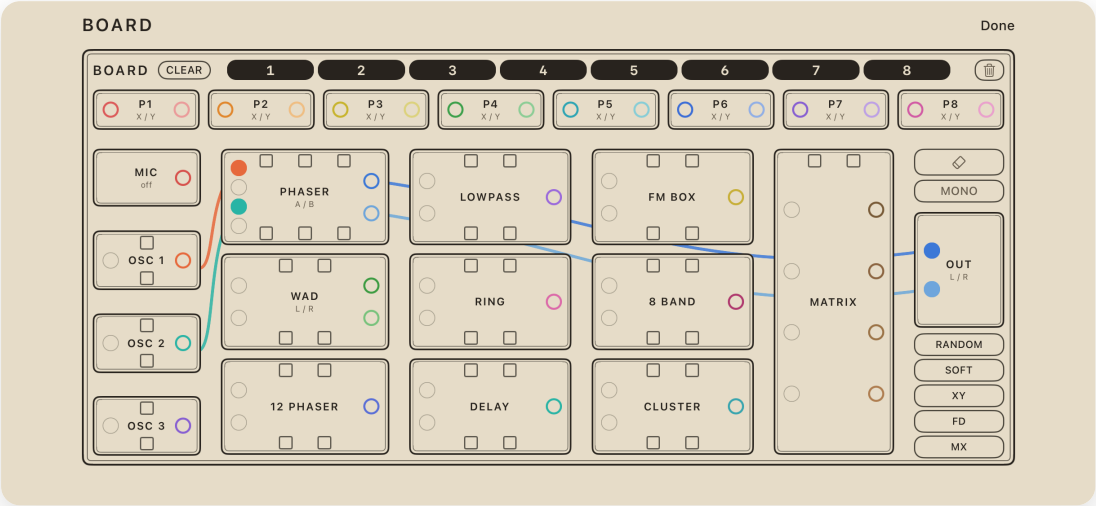
Near the edges an orbit can collapse onto a still point; when it does, the oscillator kicks itself after 48 still steps so it never stays silent.

The chip beside the name switches speed: **OSC** runs the map in the audio range, **LFO** in seconds, so any map can be a voice or a slow hand on the board. **MOD** folds the *state* of another oscillator (chosen on the chip) into this one, by the amount beside it — two maps share one orbit and drag each other about. **FM** is how much the FREQ cv on the board bends the rate.

The window draws what the map is doing — a return map, a ring or an attractor — in the colour of that oscillator's cable on the board.

05 · BOARD — CABLES, PADS AND PRESETS

Patch it with cables BOARD



Boxes and cables

On the left the sources — **MIC** and **OSC 1–3**. In the middle the phasers, **WAD** and the pedal boxes; on the right the **MATRIX** and **OUT**. Tap an output jack, then an input jack, and a cable appears. Cables are on or off; the amount is the LEVEL of the box they go into. Every box has two audio inputs, so a return can sit beside the feed, and loops — even a box into itself — are allowed. The eraser removes cables; tap a box's name to open its panel alone.

Pads and cv

The pads run along the top as **P1–P8**, X out on the left, Y on the right. Round jacks on the sides of a box are audio; the square ones above and below are cv. Only the pads send cv, so a cv cable is always thin and in its pad's colour — you can see which finger moves what.

RANDOM and its friends

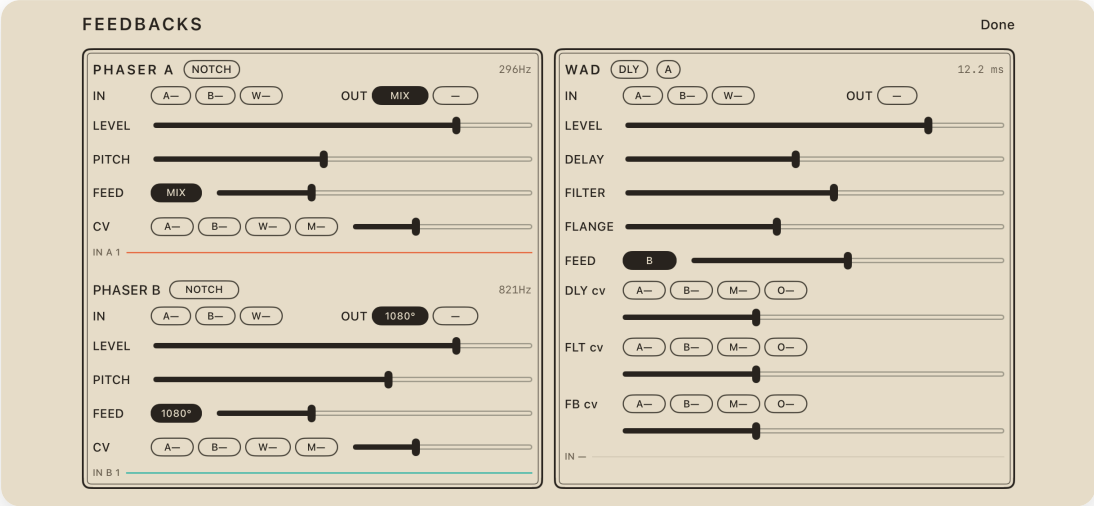
None of them draws a random number; each counts its presses and steps a fixed sequence. **RANDOM** clears the board and builds two chains into OUT left and right, with a few pad cables. **SOFT** keeps the cables and shakes every knob and switch inside. **XY** finds the circuit reaching OUT now and patches pads into its cv. **FD** adds returns to it, mostly through the matrix. **MX** rebuilds the patch so everything reaches OUT through the matrix. **MONO** sends one signal to both sides.

Presets

1–8 along the top. Tap to recall, hold to save. The whole instrument goes in: cables, matrix, every knob and switch, the pads, the oscillators, the exit EQ and the camera settings. **CLEAR** empties the board.

06 · FEEDBACKS — TWO PHASERS AND WAD

Feedback that sings FEEDBACKS



PHASER A · B

Six all-pass stages swept together, with three taps brought out — 360, 720 and 1080 degrees — and a **MIX** that adds the input back so the notches appear. **PITCH** is where the sweep sits, 20 Hz to 8 kHz. **FEED** returns one of the taps (chosen on the chip) to the input; around 0.75 the loop starts to sing, and because several frequencies cross the edge at once, it turns into a dense noise rather than one note. **NOTCH / PEAK** swaps notches for peaks.

The chips choose sources: **A** and **B** are the phasers' taps, **W** the WAD outputs, **M** the matrix outputs. Patch A into B and B into A and the stage count doubles.

WAD

A voltage-controlled, analogue-style delay. There is no clock knob: **DELAY** is a voltage, 0.5 ms to 0.6 s, and moving it stretches what is in the line so the pitch bends — that is the instrument. Output A is the delay; B is exactly twice as long and a little dirtier; **FLANGE** mixes in a very short tap; one **FILTER** sits on the way in and out.

FEED takes A, B or AUX (the dry input); zero is in the middle, left of centre returns it inverted. **DLY cv**, **FLT cv** and **FB cv** each take a source — A, B, M or **O** (the oscillators). The chip by the name (DLY / OFF / IN) chooses where the second input's cv goes.

07 · PEDALS

Small boxes with cv on the back

The pedal boxes on the board — **12 PHASER**, **LOWPASS**, **RING**, **DELAY**, **FM BOX**, **8 BAND**, **CLUSTER** — are effect *types* of the kind that came as small analogue boxes with cv jacks on the back. Each is written from how such a circuit behaves, not copied from any product, and no maker's name appears in the app. Every box has four cv jacks — two on top, two below — and a **DRIVE** at the input that clips the way a discrete input stage does. The first cv jack drives the main knob directly; the

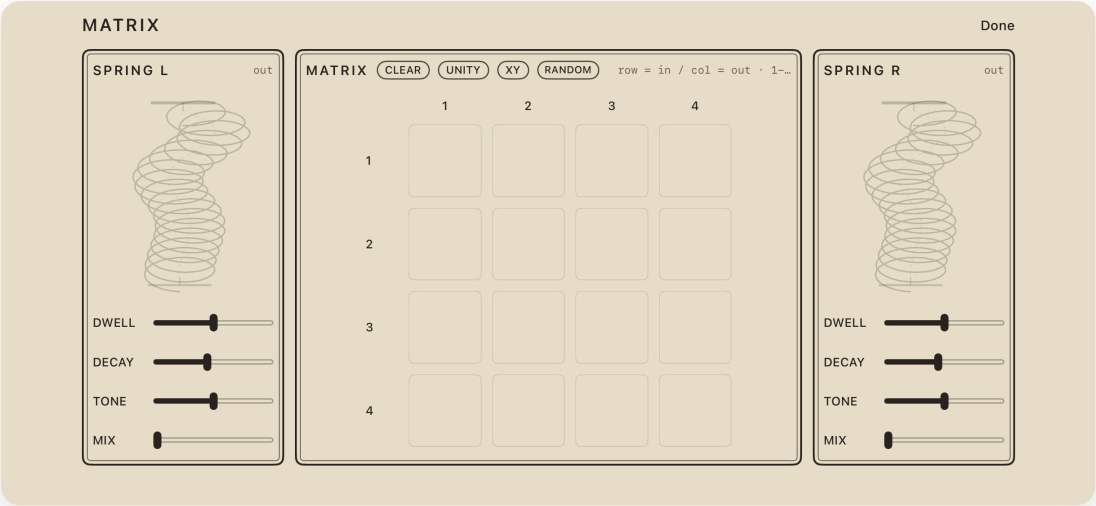
other three add up to half a knob's travel either way to the knob they are named for.

Tap a box on the board to open just that pedal.

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08 · MATRIX — AND THE TWO SPRINGS

Where the paths cross MATRIX



MATRIX

A 4×4 matrix mixer: rows are its four inputs on the board, columns its four outputs. Drag a cell up or down for the amount, or tap it to step through 0, ¼, ½, ¾, 1. **UNITY** sets the diagonal, **CLEAR** empties it, **RANDOM** fills it. Two cv jacks on the box scale the output pairs 1–2 and 3–4.

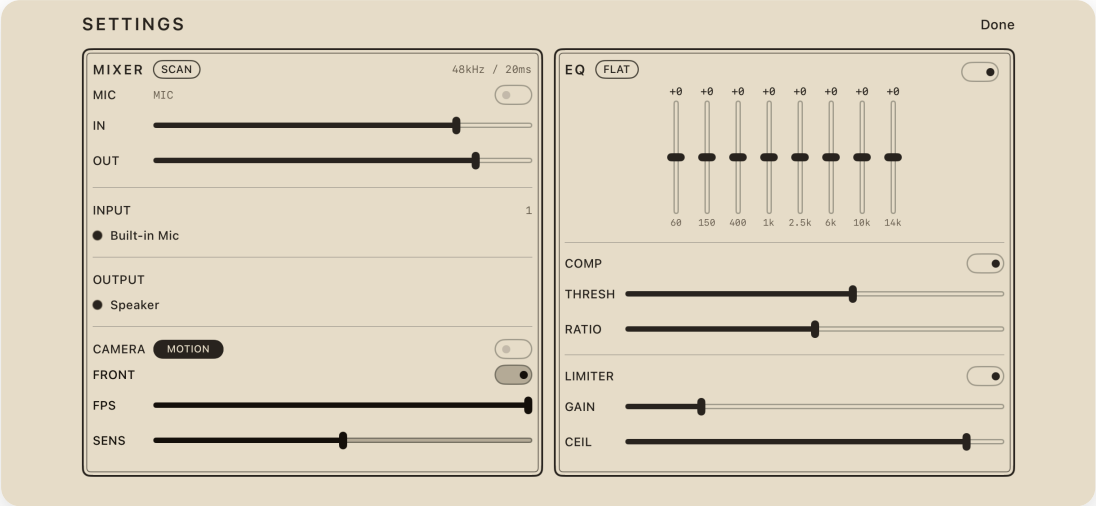
XY is a separate circuit: the sixteen cells are held by the eight pads directly — X for the even cells, Y for the odd — so the front of the instrument becomes the mixer.

SPRING L · R

Two spring reverbs, one on each side of the output: **DWELL** is how hard the springs are driven, **DECAY** how long they ring, **TONE** how bright, **MIX** how much is heard. The drawing shakes with the sound.

09 · SETTINGS — THE EXIT DESK AND THE CAMERA

The exit, and the room SETTINGS



MIXER

The microphone switch, with EX or MIC beside it — an external interface takes over the moment it is plugged in. **IN** and **OUT** levels, the inputs to pick from, and the output iOS chose. **SCAN** looks again.

CAMERA

While camera mode is on, each pad becomes a small mosaic of what the camera sees and its pointer is drawn toward the cell that is moving most — or brightest, or darkest, by the mode chip. Fingers are ignored while it runs; the pad's frame breathes with the motion, so the board's cv is visibly being played by the room. **FRONT** picks the camera, **FPS** its rate, **SENS** how much motion counts.

EQ · COMP · LIMITER

An eight-band **EQ** from 60 Hz to 14 kHz, ±12 dB (**FLAT** resets it); a compressor with **THRESH** and **RATIO**; a limiter with make-up **GAIN** and a **CEIL**. Each has its own switch. The order is master level, EQ, compressor, limiter, out.

10 · VERSIONS

Where it is, and what comes next

CURRENT

1.1.2

iPhone, landscape · iOS 18.0 or later. This guide describes this version.

NEXT UPDATE

1.2.0

iPhone and iPad, landscape · iOS 18.0 or later. In preparation.

Changes in 1.2.0

iPad	Þunresdæg runs on iPad, in landscape and full screen. The whole instrument fills the display, so the pads are larger.
Panels on iPad	Every panel opens full screen with the same layout as on iPhone, scaled up to fill it.

Details may change before release. The update will be announced in the newsletter.

Written in Swift with AVAudioEngine, with no third-party audio code. Þunresdæg, Old English for Thursday.

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